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Knitted upper with two sides and an underfoot portion

Abstract

The present disclosure provides a knitted component for an article of footwear. The knitted component may include a first side and a second side, where the first side and the second side form an overfoot portion. An underfoot portion may be located between the first side and the second side, where at least a portion of the first side and at least a portion of the second side of the knitted component are located on one of a medial side and a lateral side of the overfoot portion, and where a knitted course extends from the first side, through the underfoot portion, and to the second side.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
474574	12/1891	Bruzon	N/A	N/A
660284	12/1899	George	N/A	N/A
1521338	12/1923	Swanson	N/A	N/A
1745654	12/1929	Nealy	N/A	N/A
2101723	12/1936	Elmer	N/A	N/A
2109751	12/1937	Eugen et al.	N/A	N/A
2302596	12/1941	Albert	N/A	N/A
2677825	12/1953	Richardson et al.	N/A	N/A
2891328	12/1958	Joseph	N/A	N/A
2934838	12/1959	Ferreira	N/A	N/A
3650051	12/1971	Sass	36/133	A43B 5/02
3703775	12/1971	Gatti	N/A	N/A
3762075	12/1972	Munschy	N/A	N/A
4422249	12/1982	Hannah	N/A	N/A
4517753	12/1984	Rosenbaum et al.	N/A	N/A
4616432	12/1985	Bunch et al.	N/A	N/A
4617746	12/1985	Hannah	N/A	N/A
5020247	12/1990	Barret et al.	N/A	N/A
5106445	12/1991	Fukuoka	N/A	N/A
5167084	12/1991	Flammier	N/A	N/A
6449879	12/2001	Fallon	36/50.1	A43B 5/02
6757991	12/2003	Sussmann	N/A	N/A
7043788	12/2005	Sheets et al.	N/A	N/A
7127837	12/2005	Ito	N/A	N/A
7363731	12/2007	Dana, III	N/A	N/A
7543397	12/2008	Kilgore et al.	N/A	N/A
7913424	12/2010	Raith et al.	N/A	N/A
7954259	12/2010	Antonelli et al.	N/A	N/A

8196322	12/2011	Atsumi et al.	N/A	N/A
8272148	12/2011	Nishiwaki et al.	N/A	N/A
8347438	12/2012	Bell	N/A	N/A
8595878	12/2012	Huffa et al.	N/A	N/A
8631590	12/2013	Droege et al.	N/A	N/A
8844171	12/2013	Eder et al.	N/A	N/A
8973410	12/2014	Podhajny	36/47	D04B 7/04
8997529	12/2014	Podhajny	36/47	A43C 1/04
9038288	12/2014	Droege et al.	N/A	N/A
9179732	12/2014	Minami	N/A	N/A
9192204	12/2014	Liles et al.	N/A	N/A
9375046	12/2015	Meir	N/A	D04B 1/22
9549591	12/2016	Uchikawa	N/A	D04B 1/108
9609904	12/2016	Zwick et al.	N/A	N/A
9637847	12/2016	Terai et al.	N/A	N/A
9661892	12/2016	Meir	N/A	A43B 23/0265
9743710	12/2016	Madelaine et al.	N/A	N/A
9771673	12/2016	Ikenaka	N/A	A43B 1/04
9775401	12/2016	Cross	N/A	D04B 1/16
9907349	12/2017	Huffman	N/A	A43B 1/04
9936757	12/2017	Podhajny	N/A	A43B 23/042
9976236	12/2017	Terai	N/A	D04B 1/24
10136695	12/2017	Smith et al.	N/A	N/A
10172422	12/2018	Podhajny	N/A	A43B 1/04
10178893	12/2018	Baker	N/A	A43B 3/242
10182621	12/2018	Holt	N/A	N/A
10342288	12/2018	Yoshida et al.	N/A	N/A
10435825	12/2018	MacGilbert	N/A	D04B 7/14
10499710	12/2018	Innocente	N/A	N/A
2003/0089136	12/2002	Lynch	66/187	D04B 9/40
2008/0086914	12/2007	Curry	N/A	N/A
2011/0197475	12/2010	Weidl et al.	N/A	N/A
2014/0202034	12/2013	Vanatta	N/A	N/A
2016/0081419	12/2015	Theoklitos et al.	N/A	N/A
2017/0000210	12/2016	Bigolin	N/A	N/A
2018/0184755	12/2017	Yumiba	N/A	A43B 9/02
2018/0199668	12/2017	Yumiba	N/A	D04B 7/30
2018/0303204	12/2017	Woodard	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2727714	12/2011	CA	N/A
101152027	12/2007	CN	N/A
101768825	12/2009	CN	N/A
202849693	12/2012	CN	N/A
105121725	12/2014	CN	N/A
205030620	12/2015	CN	N/A
106460263	12/2016	CN	N/A
205913005	12/2016	CN	N/A
0562697	12/1992	EP	N/A

2987897	12/2015	EP	N/A
2014/171300	12/2013	WO	N/A
2015038243	12/2014	WO	N/A

OTHER PUBLICATIONS

Intention to Grant received for European Patent Application No. 18723173.3, mailed on Oct. 5, 2021, 7 pages. cited by applicant

Extended European Search Report received for European Patent Application No. 22150247.9, mailed on Jun. 14, 2022, 11 pages. cited by applicant

Office Action received for European Patent Application No. 22150247.9 mailed on Feb. 24, 2023, 7 pages. cited by applicant

Office Action received for European Patent Application No. 22150247.9, mailed on Sep. 14, 2023, 5 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. application Ser. No. 15/958,513, filed Apr. 20, 2018, which claims the benefit of U.S. Provisional Application No. 62/488,659, filed Apr. 21, 2017, the entireties of which are hereby incorporated by reference.

BACKGROUND

(1) A variety of articles are formed from textiles. As examples, articles of apparel (e.g., shirts, pants, socks, footwear, jackets and other outerwear, briefs and other undergarments, hats and other headwear), containers (e.g., backpacks, bags), and upholstery for furniture (e.g., chairs, couches, car seats) are often at least partially formed from textiles. These textiles are often formed by weaving or interlooping (e.g., knitting) a yarn or a plurality of yarns, usually through a mechanical process involving looms or knitting machines. One particular object that may be formed from a textile is an upper for an article of footwear.

(2) Conventional articles of footwear generally include two primary elements: an upper and a sole structure. The upper is secured to the sole structure and forms a void within the article of footwear for comfortably and securely receiving a foot. The sole structure is secured to a lower surface of the upper so as to be positioned between the upper and the ground. In some articles of athletic footwear, for example, the sole structure may include a midsole and an outsole. The midsole may be formed from a polymer foam material that attenuates ground reaction forces to lessen stresses upon the foot and leg during walking, running, and other ambulatory activities. The outsole may be secured to a lower surface of the midsole and forms a ground-engaging portion of the sole structure that is formed from a durable and wear-resistant material.

(3) The upper of the article of footwear generally extends over the instep and toe areas of the foot, along the medial and lateral sides of the foot, and around the heel area of the foot. Access to the void on the interior of the upper is generally provided by an ankle opening in a heel area of the footwear. A lacing system is often incorporated into the upper to adjust the fit of the upper, thereby facilitating entry and removal of the foot from the void within the upper. The upper may include a

tongue that extends under the lacing system to enhance adjustability of the footwear, and the upper may incorporate a heel counter to limit movement of the heel.

BRIEF SUMMARY

(4) In one aspect, the present disclosure provides a knitted component for an article of footwear, the knitted component including a first side and a second side, the first side and the second side forming an overfoot portion, and an underfoot portion positioned between the first side and the second side. At least a portion of the first side and at least a portion of the second side of the knitted component may each be configured to be positioned on one of a medial side and a lateral side of the overfoot portion, and a course of the knitted component may extend from the first side, through the underfoot portion, and to the second side.

(5) The course may be formed by a single pass of a knitting machine.

(6) The first side may include a toe portion, the toe portion having an underfoot side, an overfoot side, and a cavity therebetween. The cavity may form at least a portion of a void for receiving a foot. The toe portion may include a connection structure formed on a knitting machine and joining the underfoot side and the overfoot side. The underfoot side of the toe portion may include a first edge, where the underfoot portion includes a second edge, and where the first edge is joined to the second edge. The first edge and the second edge may be joined at a seam, and the seam may be configured to be located beneath a foot of a wearer when the article of footwear is in use.

(7) The knitted component may further include a cup-shaped heel portion with a first edge configured to form a collar of the article of footwear. The heel portion may extend from the underfoot portion.

(8) A fastening element may be coupled to the first side and the second side and configured to pull the first side with respect to the second side to tighten the knitted component around a foot.

(9) In another aspect, the present disclosure provides a method. The method may include knitting a first side of a knitted component, knitting a second side of the knitted component, where the first side and the second side form an overfoot portion, and knitting an underfoot portion positioned between the first side and the second side. At least a portion of the first side and at least a portion of the second side of the knitted component may be located on one of a medial side and a lateral side of the overfoot portion, and a knitted course may extend from the first side, through the underfoot portion, and to the second side.

(10) The course may be formed by a single pass of a knitting machine.

(11) The first side may include a toe portion, the toe portion having an underfoot side, an overfoot side, and a cavity therebetween. Knitting the toe portion of the first side may include knitting the overfoot side while holding loops on a needle bed, and joining the loops held on the needle bed to the underfoot side when knitting the underfoot side. A connection structure may be formed when joining the loops held on the needle bed to the underfoot side, where the connection structure joins the overfoot side and the underfoot side. The cavity may form at least a portion of a void for receiving a foot. The underfoot side of the toe portion may include a first edge, where the underfoot portion includes a second edge, and where the first edge is joined to the second edge. The method may further include joining the first edge and the second edge at a seam, where the seam is configured to be located beneath a foot of a wearer when the knitted component is incorporated into an article of footwear.

(12) The knitted component may include a cup-shaped heel portion with a first edge configured to form a collar defining an opening to a void. Knitting the heel portion may include knitting a first portion of the heel portion while holding loops on a needle bed, and joining the loops held on the needle bed to a second portion of the heel portion while knitting the second portion of the heel portion, where joining the loops held on the needle bed to the second portion forms a connection structure joining the first portion to the second portion.

(13) In another aspect, the present disclosure provides another embodiment of a knitted component for an article of footwear. The knitted component may include a first side and a second side, the

first side and the second side forming an overfoot portion. The knitted component may further include an underfoot portion positioned between the first side and the second side, where the first side includes a toe portion, the toe portion having an underfoot side, an overfoot side, and a cavity therebetween, where the cavity forms at least a portion of a void for receiving a foot, and where the toe portion includes a connection structure formed on a knitting machine and joining the underfoot side and the overfoot side.

(14) A loop included in a course forming the overfoot side may be held on a needle bed when forming the overfoot side, and the loop may be included in the connection structure.

(15) In another aspect, the present disclosure provides another embodiment of a method. The method may include knitting a first side of a knitted component, where the first side includes a toe portion, the toe portion having an underfoot side, an overfoot side, and a cavity therebetween, where the cavity forms at least a portion of a void for receiving a foot, and where knitting the toe portion includes knitting a connection structure joining the underfoot side and the overfoot side.

(16) The connection structure may include holding a loop on a needle bed of the knitting machine for at least one course. Knitting the connection structure may include holding a loop on a needle bed while knitting at least a portion of the overfoot portion, and then knitting with the loop held on the needle bed when knitting the underfoot portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration showing an article of footwear formed with a knitted component in accordance with an embodiment of the present disclosure.

(2) FIG. 2 is an illustration showing a top view of a knitted component with an overfoot portion and an underfoot portion included in the article of footwear of FIG. 1, where the knitted component is in an unfolded state.

(3) FIG. 3 is an illustration showing a top view of the knitted component of FIG. 2 in a folded state.

(4) FIG. 4 is an illustration showing a bottom view of the knitted component of FIGS. 2-3 in the unfolded state.

(5) FIG. 5 is an illustration showing a bottom view of the knitted component of FIGS. 2-4 in the folded state.

(6) FIG. 6 is an illustration showing a lateral-side view of the knitted component of FIGS. 2-5 in the folded state.

(7) FIG. 7 is an illustration showing a medial-side view of the knitted component of FIGS. 2-6 in the folded state.

(8) FIG. 8 is an illustration showing the knitted component of FIGS. 2-7 when being formed on a knitting machine.

(9) FIG. 9 is an illustration showing a close-up view of a toe portion of the knitted component of FIGS. 2-8.

(10) FIG. 10 is a diagram of an embodiment of a knitting process for forming a portion of the knitted component of FIG. 9.

(11) FIG. 11 is a diagram of an embodiment of a knitting process for forming the knitted component of FIGS. 2-9.

DETAILED DESCRIPTION

(12) Various aspects are described below with reference to the drawings in which like elements generally are identified by like numerals. The relationship and functioning of the various elements of the aspects may better be understood by reference to the following detailed description.

However, aspects are not limited to those illustrated in the drawings or explicitly described below.

It also should be understood that the drawings are not necessarily to scale, and in certain instances details may have been omitted that are not necessary for an understanding of aspects disclosed herein, such as conventional fabrication and assembly.

(13) Certain aspects of the present disclosure relate to articles at least partially formed from textiles. One example of an article is an article of apparel (e.g., shirts, pants, socks, footwear, jackets and other outerwear, briefs and other undergarments, hats and other headwear, or the like). The article may be an upper configured for use in an article of footwear. The upper may be used in connection with any type of footwear. Illustrative, non-limiting examples of articles of footwear include a basketball shoe, a biking shoe, a cross-training shoe, a global football (soccer) shoe, an American football shoe, a bowling shoe, a golf shoe, a hiking shoe, a ski or snowboarding boot, a tennis shoe, a running shoe, and a walking shoe. The upper may also be incorporated into a non-athletic shoe, such as a dress shoe, a loafer, and a sandal.

(14) Referring to FIG. 1, an article of footwear **100** may include an upper **102** secured to a sole structure **104**. The upper **102** may include a lateral side **106** and a medial side **108**. The area where the sole structure **104** joins the upper **102** may be referred to as the biteline **110**. The upper **102** may be joined to the sole structure **104** in a fixed manner using any suitable technique, such as through the use of an adhesive, by sewing, etc. It is contemplated that the upper **102** may extend partially or completely around a foot **112** of a wearer and/or may be integral with the sole, and a sockliner may or may not be used. In some embodiments, the sole structure **104** may include a midsole (not shown) and an outsole.

(15) The article of footwear **100** may additionally comprise a throat area **114** and an ankle opening **116**, which may be surrounded by a collar **118** and may lead to a void **120**. The void **120** of the article of footwear **100** may be configured to accommodate a foot of a person. The throat area **114** may be generally disposed in a midfoot area **122** of the upper **102**. The midfoot area **122** of the upper **102** may be located between a heel area **124** and a toe area **126**. In some embodiments, a tongue may be disposed in the throat area **114**, but a tongue is an optional component. The tongue may be any type of tongue, such as a gusseted tongue or a burrito tongue. If a tongue is not included (or in combination with a tongue), the lateral and medial sides of the throat area **114** may be joined together. For example, as depicted in FIG. 1 and described in more detail below, the upper **102** may be formed into its wearable shape when a first side **128** is joined to and/or overlaps a second side **130**.

(16) As shown in FIG. 1, the article of footwear **100** may include an optional fastening element **132** (which may be associated with the apertures **133**). Any suitable type of fastening element may be used, such as a shoelace, cable-tensioning system, and/or any other suitable device. The fastening element **132** may be coupled to both the first side **128** and the second side **130** of the upper **102** and may pull the first side **128** and the second side **130** together in response to actuation initiated by the user or another device, for example, to thereby tighten the upper **102** around the foot **112**.

(17) Referring to FIG. 2, at least a portion of the upper **102**, and potentially substantially the entirety of the upper **102**, may be formed of a knitted component **134**. FIG. 2 shows the upper **102** formed by the knitted component **134** as it may appear after a knitting process (such as a weft-knitting process on a flat knitting machine as described in FIG. 8 below). The knitted component **134** may form an underfoot portion **136**, as shown. The underfoot portion **136** may be configured to extend under the foot of a wearer when the article of footwear is in use and may couple to, and/or at least partially form, a midsole and/or an outsole. The knitted component **134** may be formed as an integral one-piece element during a knitting process, such as a weft knitting process (e.g., with a flat knitting machine or circular knitting machine), a warp knitting process, or any other suitable knitting process. That is, the knitting process on the knitting machine may substantially form the knit structure of the knitted component **134** without the need for significant post-knitting processes or steps. Alternatively, two or more portions of the knitted component **134** may be formed separately as distinct integral one-piece elements and then the respective elements attached.

(18) Forming the upper **102** with the knitted component **134** may provide the upper **102** with advantageous characteristics including, but not limited to, a particular degree of elasticity (for example, as expressed in terms of Young's modulus), breathability, bendability, strength, moisture absorption, weight, abrasion resistance, and/or a combination thereof. These characteristics may be accomplished by selecting a particular single layer or multi-layer knit structure (e.g., a ribbed knit structure, a single jersey knit structure, or a double jersey knit structure), by varying the size and tension of the knit structure, by using one or more yarns formed of a particular material (e.g., a polyester material, a relatively inelastic material, or a relatively elastic material such as spandex), by selecting yarns of a particular size (e.g., denier), and/or a combination thereof. The knitted component **134** may also provide desirable aesthetic characteristics by incorporating yarns having different colors, textures or other visual properties arranged in a particular pattern. The yarns themselves and/or the knit structure formed by one or more of the yarns of the knitted component **134** may be varied at different locations such that the knitted component **134** has two or more portions with different properties (e.g., a portion forming the throat area **114** of the upper **102** may be relatively elastic while another portion may be relatively inelastic). In some embodiments, the knitted component **134** may incorporate one or more materials with properties that change in response to a stimulus (e.g., temperature, moisture, electrical current, magnetic field, or light). For example, the knitted component **134** may include yarns formed of a thermoplastic polymer material (e.g., polyurethanes, polyamides, polyolefins, and nylons) that transitions from a solid state to a softened or liquid state when subjected to certain temperatures at or above its melting point and then transitions back to the solid state when cooled. The thermoplastic polymer material may provide the ability to heat and then cool a portion of the knitted component **134** to thereby form an area of bonded or continuous material that exhibits certain advantageous properties including a relatively high degree of rigidity, strength, and water resistance, for example.

(19) Still referring to FIG. 2, the knitted component **134** may include the first side **128** which may extend from a medial side **108** of the underfoot portion **136**, and a second side **130** which may extend from a lateral side **106** of the underfoot portion **136** (or vice versa). The first side **128** may extend a distance from the underfoot portion **136** such that it has a first portion **138** corresponding to (e.g., configured to form a portion of) the medial side of the upper **102**, a second portion **140** corresponding to the throat area **114** of the upper **102**, and a third portion **142** corresponding to the lateral side of the upper **102**. The second side **130**, which may include less area than the area of the first side **128**, may be configured to form only a portion of the lateral side of the upper **102** (as shown). Alternatively, the second side **130** may be larger than depicted in FIG. 2 when it is desirable for the second side **130** to extend beyond the lateral side (e.g., to the throat area and/or to the medial side) when folded. Alternatively, the first side **128** and the second side **130** could be switched around with respect to the medial and lateral directions.

(20) The first side **128** may include a toe portion **144** of the knitted component **134**, which may be located in the toe area **126** of the completed upper **102** (shown in FIG. 1). As depicted in FIG. 2 and described in more detail below with reference to FIGS. 9-10, the toe portion **144** may include a cup-like shape such that an underfoot toe surface **146** (shaded for illustration) is opposite an underfoot midfoot surface **148** (and it is noted that the underfoot midfoot surface **148** is facing away from the viewer such that it is not visible in FIG. 2; see FIG. 4). Similarly, a heel portion **145** may extend from the underfoot portion **136** and have a cup-like shape for receiving a heel of the wearer. When the first side **128** is folded along the first arrow **150**, the underfoot toe surface **146** may invert such that, once the fold is complete, it faces the same direction as the underfoot midfoot surface **148**. An edge **152** of the toe portion **144** may then be joined to an edge **154** of the underfoot midfoot surface **148** as described in more detail below (with reference to FIG. 5). A rear edge **164** of the first side **128** and a rear edge **166** of the second side **130** may be configured to form a collar of the upper.

(21) When forming the upper **102**, and after the knitting process, the second side **130** may be

folded in a direction corresponding to the second arrow **156** and opposite of the direction that the first arrow **150**. Thus, when the first side **128** and/or the second side **130** extend far enough from the underfoot portion **136**, an outer edge **174** of the first side **128** and an outer edge **176** the second side **130** may overlap when the upper **102** is formed (as shown in FIG. **6**). Alternatively, it is contemplated that a different, non-depicted portion (e.g., a tongue) of the upper may be located between the first side **128** and the second side **130**. Also, while not shown, the second side **130** may extend far enough from the underfoot portion **136** such that it can reach the throat area and even the lateral side of the upper when folded, which may be advantageous when it is desirable to provide two layers of knitted material over the foot, for example.

(22) FIG. **3** shows the upper **102** of FIG. **2** in a folded state. Folding or otherwise manipulating the upper **102** into an appropriate shape for use in an article of footwear may include incorporating a last and an associated method of lasting. An example of a lasting process is described in U.S. patent application Ser. No. 12/848,352, filed Aug. 2, 2010, and issued as U.S. Pat. No. 8,595,878, which is herein incorporated by reference in its entirety.

(23) As shown in FIG. **3**, the first side **128** has been folded such that the first portion **138** forms an outer surface **160** of the upper **102** on the medial side **108**, the second portion **140** forms the outer surface **160** of the upper **102** at the throat area **114**, and the third portion **142** forms the outer surface **160** on the lateral side **106** of the upper **102**. Similarly, the second side **130** forms the outer surface **160** on the lateral side of the upper **102**. Further, the first side **128** and the second side **130** overlap at the overlapped area **162**. Either of the first side **128** and the second side **130** may be exposed on the outer surface **160** at the overlap area **162** (and in this case, the first side **128**). The rear edge **164** of the first side **128** and the rear edge **166** of the second side may form at least a portion of the collar **118**, as shown. In the toe area **126**, the outer surface **160** may be formed by the toe portion **144** (and specifically an overfoot toe surface **168** of the toe portion **144**), which may be a portion of the first side **128**.

(24) FIG. **4** shows a bottom view of the upper **102**, again in the unfolded state of FIG. **2**. The underfoot midfoot surface **148** of the underfoot portion **136**, which is positioned between the first side **128** and the second side **130**, is shaded for illustration, but this may or may not be the case in practice. It is contemplated that, prior to folding or otherwise manipulating the upper **102** into its wearable shape, the first side **128** and/or the second side **130** may not be discernable from the underfoot portion **136**.

(25) As shown in FIG. **4**, the toe portion **144** of the first side **128** may include an overfoot toe surface **168** facing the same direction as the underfoot midfoot surface **148** in the depicted unfolded state. Once the upper **102** is manipulated into its folded state, the overfoot toe surface **168** may invert such that it is located above a toe of the wearer in the completed upper **102**. The underfoot toe surface **146** (described above and depicted in FIG. **2**) is facing away from the viewer in FIG. **4**.

(26) FIG. **5** shows the bottom view of FIG. **4** when the upper **102** is manipulated to its folded state. As shown, the underfoot midfoot surface **148** remains facing away from the foot of a wearer and towards the viewer in FIG. **5**. The underfoot toe surface **146** has been inverted with respect to the unfolded state such that, in the folded state, the underfoot toe surface **146** faces the same direction as the underfoot midfoot surface **148**. The overfoot toe surface **168** (of FIG. **4**) has also been inverted and now faces away from the viewer, and thus it is not visible in FIG. **5**.

(27) Once in the folded state, the edge **152** of the toe portion **144** may then be aligned with and adjacent to the edge **154** of the underfoot midfoot surface **148**. In some embodiments, the edge **152** may be joined to the edge **154** by any suitable apparatus or method, and it is contemplated that the step of joining the edge **152** to the edge **154** may take place after the knitted component **134** is removed from the knitting machine and during or after the lasting step. Joining the edge **152** to the edge **154** may include sewing or tying, using an adhesive, using a mechanical clamp, etc. In some embodiments, the edge **152** and the edge **154** may be indirectly coupled (i.e., due to each independently being joined to a sole structure).

(28) FIG. 6 shows another view (i.e., a lateral-side view) of the upper **102** in the folded state and on a last **158**. As shown, the edge **152** and the edge **154** may form a seam **170** located in the toe area of the upper **102** and under the foot of a wearer. Advantageously, the current embodiment may hide the seam from view (e.g., when a sole structure covers the seam), thus potentially increasing the aesthetic appeal of the article of footwear and protecting the seam **170** from wear and tear. The seam **170** may have an end **172** where the first side **128** and the second side **130** meet when the upper **102** is in the folded state. As shown, the first side **128** and the second side **130** may optionally overlap at the overlap area **162**, and the overlapped area **162** may terminate at the end **172** of the seam **170**.

(29) FIG. 7 shows the upper **102** (without hidden lines) shown from a medial-side perspective when in the folded state. As shown, the medial side may be substantially formed by the first side **128**. This may be advantageous where it is desirable for the medial side to have a continuous and uniform appearance, for example. Further, the first side **128** may provide the medial side **108** with particular mechanical properties (e.g., stretchability, strength, etc.) and/or uniform surface characteristics such that the upper **102** is suitable for a particular activity. For example, the medial side **108** may be formed with particular mechanical properties and particular surface characteristics for kicking a ball with the medial side of the foot, which may be a common form of kicking when playing global football (also called soccer).

(30) Referring to FIG. 8, the knitted component **134** may be formed on a knitting machine **178**. In particular, the knitted component **134** may include a seamless portion formed on the knitting machine **178** and extending from the outer edge **174** of the first side **128**, through the underfoot portion **136**, and to the outer edge **176** of the second side **130**. The outer edges **174**, **176** may be terminal ends of the knitted component **134** during knitting process and/or after the knitting process when the knitted component **134** is removed from the knitting machine **178**. When forming the knitted component **134**, the knitted component **134** may be oriented with respect to needle bed(s) **180** of the knitting machine **178** such that a feeder **182** of the knitting machine **178** is capable of moving in a single pass (i.e., without changing its feed direction) to knit a first course **184** extending from the outer edge **174** of the first side **128**, through the underfoot portion **136**, and to the outer edge **176** of the second side (and/or vice versa). Thus, when the knitted component **134** is formed and removed from the knitting machine, the continuous first course **184** (which may include one or more yarns dispensed from the feeder **182** during the single pass) may extend from the outer edge **174** of the first side **128**, through the underfoot portion **136**, and to the outer edge **176** of the second side **130**.

(31) In some embodiments, the first course **184** may include a continuous strand of yarn. Additionally or alternatively, one or more strands of yarn forming at least a portion of the first course **184** may extend less than the full length of the first course **184**. For example, it is contemplated that a strand of yarn may extend from the outer edge **174** and towards the underfoot portion **136**, but may terminate within the first course **184** prior to extending into the underfoot portion **136**. Similarly, it is contemplated that a strand of yarn may extend within the first course **184** from the underfoot portion **136** to the outer edge **176** of the second side **130**. Advantageously, the first course **184** may provide different portions of the knitted component with different characteristics (e.g., the underfoot portion **136** may have yarns suitable for attaching to a sole structure, while the yarns of the first and/or second sides **128**, **130** may be suitable for forming the overfoot portion of the upper **102**).

(32) FIG. 9 is an illustration showing a close-up view of the toe portion **144** of the first side **128** of the knitted component **134**. An underfoot side **186** of the toe portion **144** (which is inverted in FIG. 9) may form the underfoot toe surface **146** as described in more detail above, and an overfoot side **188** of the toe portion **144** may be inverted during the folding step such that it at least partially forms the toe area of the overfoot portion of the article of footwear (see FIG. 1). The underfoot side **186** and the overfoot side **188** may be curved such they form a cup-like shape, and a cavity **190**

may be formed between the underfoot side **186** and the overfoot side **188** and may form a portion of the void of the article of footwear. A connection structure **192** may secure the underfoot side **186** to the overfoot side **188**. The connection structure **192** may be formed with the rest of the knitted component (e.g., an integral one-piece element). That is, the same knitting process may form at least one of the underfoot side **186** and the overfoot side **188** on the knitting machine, and the knitting machine may also form the connection structure **192** without the need for significant post-knitting processes or steps. In alternative embodiments, the connection structure **192** may be formed after the knitting process (e.g., by sewing the underfoot side **186** and the overfoot side **188** together). While not shown in FIG. **9**, the heel portion **145** (shown in FIG. **2**) of the knitted component may have a similar cup-like structure and may be formed with a similar process.

(33) FIG. **10** is a diagram (“the knit diagram”) illustrating a knitting sequence of one method of forming the portion of the knitted component **134** depicted in FIG. **9**. Each shaded horizontal line **194** may represent a course or a certain number of courses, for example. The sequence may be performed in the direction depicted by the large bolded arrows. The labeled locations of FIG. **10** correspond with the labeled elements of FIG. **9**.

(34) Knitting courses at the central portion **196** may correspond with knitting courses extending across the first side **128**, the underfoot portion **136**, and the second side **130** of the upper **102** (and it is noted that FIG. **8** depicts a knitting machine at the time it is forming the central portion **196** of the upper **102**). Once the knitting machine reaches the edge **154**, the knitting machine may stop knitting the underfoot portion **136** and the second side **130**. It may, however, continue knitting the toe portion **144** of the first side **128**.

(35) When knitting the toe portion **144** of the first side **128**, the knitting machine may first knit the overfoot side **188** of the toe portion **144** (which is inverted when it is removed from the knitting machine). The overfoot side **188**, as it is initially formed, may be continuous with the remainder of the first side **128**. As the knitting machine continues along the overfoot side **188** and as its pattern narrows, it may hold loops forming the terminal outer portions of the overfoot side **188** on the needles of a needle bed rather than releasing them. Once the knitting machine reaches the narrow **200** (which represents the terminal end of the toe portion **144** once the knitting process is finished), the knitting machine may continue knitting the inverted underfoot side **186** while simultaneously re-joining the loops held on the needle bed as the pattern widens. The rejoining of the loops is depicted by the arrows **202**. The rejoining of the loops may form the connection structure **192** described with reference to FIG. **9** above. The end result of this process may be the cup-like toe portion **144** of the knitted component **134**, which is described in detail above. The last course formed on the knitting machine may correspond with the edge **152** (also shown in FIG. **2** and FIG. **5**).

(36) FIG. **11** is a diagram illustrating a knitting sequence of the entirety, or substantially the entirety, of the knitted component **134**. As shown, the cup-like shape of the heel portion **145** may be formed in a manner similar to the cup-like shape of the toe portion **144**. A first edge **204** may represent a first course of the knitted component formed on a knitting machine, and the first edge **204** may form a rear (e.g., heel-side) portion of the collar **118** (shown in FIG. **1**). Then, the sequence may continue in the direction depicted by the large bolded arrows. As the pattern narrows, the outer loops of the respective courses may be held on a needle bed of the knitting machine, and they may be rejoined when the pattern again widens.

(37) The central portion **196**, which may include at least a portion of the first side **128**, the second side **130**, and the underfoot portion **136**, may be substantially flat during and after the knitting process. The central portion **196** may utilize one or both beds of the knitting machine. The flat nature of the central portion **196** and the ability to utilize two beds of a flat knitting machine may enhance the ability to include certain knit or non-knit features, particularly since utilizing two beds (instead of one) significantly increases the ability to selectively include visual designs, selectively vary physical properties, and incorporate other features formed by a particular knit structure,

particular yarn(s), or a combination thereof. To illustrate, one yarn type (e.g., a thermoplastic polymer material yarn) may be located on one surface of the central portion **196** and a different yarn type (e.g., a polyester yarn) may be located on an opposite facing surface of the central portion **196**, which may be accomplished when using two needle beds. In another embodiment, the central portion **196** may optionally include separable layers (e.g., a first layer formed on a first bed, a second layer formed on a second bed, and a pocket therebetween), but separable layers are not required in all embodiments, and they are not depicted. Separable layers may be advantageous when forming a pocket for holding certain components, such as a cushioning element, for example. The present embodiments may also be advantageous for providing the ability for a single yarn type (e.g., a high tenacity yarn) to extend substantially around the foot while inlaid within the knitted component **134**.

(38) The flat nature of the central portion **196** may additionally or alternatively be advantageous for facilitating performance of post-knitting processes. For example, the flat knitted surfaces of the central portion **196** may be quickly and efficiently printed, painted, or otherwise modified such that the article of footwear includes certain visual designs (e.g., abstract patterns, logos, images, etc.). Further, the flat knitted surfaces may be treated (e.g., sprayed, painted, or printed) after the knitting process to add certain physical properties into the article of footwear (e.g., waterproof-related properties, varying elasticity, etc.).

(39) In the present disclosure, the ranges given either in absolute terms or in approximate terms are intended to encompass both, and any definitions used herein are intended to be clarifying and not limiting. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the present embodiments are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements. Moreover, all ranges disclosed herein are to be understood to encompass any and all subranges (including all fractional and whole values) subsumed therein.

(40) Furthermore, the present disclosure encompasses any and all possible combinations of some or all of the various aspects described herein. It should also be understood that various changes and modifications to the aspects described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present disclosure and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A method of forming an upper of an article of footwear, the method comprising: knitting a first side of a knitted component; knitting a second side of the knitted component, wherein the first side and the second side comprise an overfoot portion; knitting an underfoot portion located between the first side and the second side; knitting a toe portion on the first side of the knitted component, wherein the toe portion has an underfoot side, an overfoot side, and a cavity between the underfoot side and the overfoot side; knitting the overfoot side of the toe portion while holding loops on a needle bed of a knitting machine, and joining the loops held on the needle bed to the underfoot side when knitting the underfoot side of the toe portion, wherein a connection structure is formed when joining the loops held on the needle bed to the underfoot side, and wherein the connection structure joins the overfoot side and the underfoot side of the toe portion; wherein the underfoot side of the toe portion includes a first edge, wherein the underfoot portion includes a second edge, and wherein the method includes joining the first edge to the second edge; wherein at least a portion of the first side and at least a portion of the second side of the knitted component are both located on one of a medial side or a lateral side of the overfoot portion, wherein when the first side and the second side are joined to form the overfoot portion, the first side is secured to the second side along

an overlapped area on the one of the medial side or the lateral side, the overlapped area extending from a toe area to an ankle opening, and wherein a knitted course extends from the first side, through the underfoot portion, and to the second side.

2. The method of claim 1, wherein a connection structure connects the first edge and the second edge together.

3. The method of claim 1, wherein the knitted component is knitted to include a cup-shaped heel portion.

4. The method of claim 1, wherein a first strand of yarn within the knitted course extends from the first side and terminates within the underfoot portion.

5. The method of claim 1, further comprising manipulating the knitted component into a dimensional shape of the upper.

6. The method of claim 1, further comprising concealing the seam.

7. The method of claim 1, wherein the first edge and the second edge are joined together via sewing.

8. The method of claim 1, wherein the first edge and the second edge are joined together via an adhesive.

9. A method of forming an upper of an article of footwear, the method comprising: knitting a first side of a knitted component; knitting a second side of the knitted component, wherein the first side and the second side are configured to overlap and wherein the first side and the second side collectively define an overfoot portion; knitting an underfoot portion located between the first side and the second side, the underfoot portion comprising a first edge and a second edge; knitting a toe portion on the first side of the knitted component, the toe portion comprising an underfoot side, an overfoot side, and a cavity located between the underfoot side and the overfoot side; knitting the overfoot side of the toe portion while holding loops on a needle bed of a knitting machine; joining the loops held on the needle bed to the underfoot side of the toe portion while knitting the underfoot side, thereby forming a connection structure that connects the overfoot side to the underfoot side; manipulating the knitted component into a dimensional shape of the upper; and joining the first edge and the second edge of the underfoot portion together along an overlapped area on a medial side or a lateral side of the overfoot portion, the overlapped area extending from a toe region to an ankle opening; and knitting a knitted course that extends continuously from the first side, through the underfoot portion, and to the second side.

10. The method of claim 9, wherein the first edge comprises part of a boundary of a toe portion of the knitted component.

11. The method of claim 9, further comprising joining the upper to a sole structure, wherein an entire length of the seam is covered by the sole structure.

12. The method of claim 9, wherein the first edge and the second edge are joined together via sewing.

13. The method of claim 9, wherein the first edge and the second edge are joined together via an adhesive.

14. A method, comprising: knitting a first side of a knitted component; knitting a second side of the knitted component, wherein the first side and the second side comprise an overfoot portion; knitting an underfoot portion located between the first side and the second side; knitting a toe portion on the first side of the knitted component, wherein the toe portion has an underfoot side, an overfoot side, and a cavity between the underfoot side and the overfoot side; knitting the overfoot side of the toe portion while holding loops on a needle bed of a knitting machine, and joining the loops held on the needle bed to the underfoot side when knitting the underfoot side of the toe portion, wherein a connection structure is formed when joining the loops held on the needle bed to the underfoot side, and wherein the connection structure joins the overfoot side and the underfoot side of the toe portion; wherein the underfoot side of the toe portion includes a first edge, wherein the underfoot portion includes a second edge, and wherein the method includes joining the first

edge to the second edge; wherein at least a portion of the first side and at least a portion of the second side of the knitted component are both located on one of a medial side or a lateral side of the overfoot portion, wherein when the first side and the second side are joined to form the overfoot portion, the first side is secured to the second side along an overlapped area on the one of the medial side or the lateral side, the overlapped area extending from a toe area to an ankle opening, and wherein a knitted course extends from the first side, through the underfoot portion, and to the second side.
